

Customer Shopping Behavior Analysis

1. Project Overview

This project analyzes customer shopping behavior using transactional data from 3,900 purchases across various product categories. The goal is to uncover insights into spending patterns, customer segments, product preferences, and subscription behavior to guide strategic business decisions.

2. Dataset Summary

- Rows: 3,900
- Columns: 19
- Key Features:
 - Customer demographics (Age, Gender, Location, Subscription Status)
 - Purchase details (Item Purchased, Category, Purchase Amount, Season, Size, Color)
 - Shopping behavior (Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type)
- Missing Data: None found in the raw file; all 19 columns are fully populated across all 3,900 records.

3. Exploratory Data Analysis using Python

We began with data preparation and cleaning in Python:

- **Data Loading:** Imported the dataset using `pandas`.
- **Initial Exploration:** Used `df.info()` to check structure and `.describe()` for summary statistics.
- **Missing Data Handling:** Confirmed there were no null values in any column, so no imputation was required.
- **Column Standardization:** Renamed columns to snake_case for better readability and documentation.
- **Feature Engineering:**
 - Created `age_group` column by binning customer ages into quartiles (Young Adult, Adult, Middle-aged, Senior).
 - Created `customer_segment` column by binning customers on previous purchase counts (New, Returning, Loyal).
- **Data Consistency Check:** Compared `discount_applied` and `promo_code_used` for redundancy and kept both, since they did not align on every record.
- **Database Integration:** Loaded the cleaned DataFrame into a structured table for SQL-style analysis.

4. Data Analysis using SQL (Business Transactions)

We performed structured analysis to answer key business questions:

1. **Revenue by Gender** – Compared total revenue generated by male vs. female customers.

Gender	Revenue (USD)
Male	157,890
Female	75,191

2. **High-Spending Discount Users** – Identified customers who used discounts but still spent above the average purchase amount (\$59.76).

Metric	Value
Customers meeting criteria	839
Share of total customers	21.5%

3. **Top 5 Products by Rating** – Found products with the highest average review ratings.

Item Purchased	Average Rating
Gloves	3.86
Sandals	3.84
Boots	3.81
Hat	3.81
T-shirt	3.78

4. **Shipping Type Comparison** – Compared average purchase amounts across shipping types.

Shipping Type	Avg. Purchase Amount
2-Day Shipping	\$60.73
Express	\$60.48
Free Shipping	\$60.41
Store Pickup	\$59.89
Next Day Air	\$58.63
Standard	\$58.46

5. **Subscribers vs. Non-Subscribers** – Compared average spend and total revenue across subscription status.

Subscription Status	Total Customers	Avg. Spend	Total Revenue
No	2,847	\$59.87	\$170,436
Yes	1,053	\$59.49	\$62,645

6. **Discount-Dependent Products** – Identified the products with the highest percentage of discounted purchases.

Item Purchased	Discount Rate (%)
Hat	50.00
Sneakers	49.66
Coat	49.07
Sweater	48.17
Pants	47.37

7. **Customer Segmentation** – Classified customers into New, Returning, and Loyal segments based on previous purchase counts.

Customer Segment	Number of Customers
Loyal (>10 previous purchases)	3,116
Returning (2–10 previous purchases)	701
New (1 previous purchase)	83

8. **Top 3 Products per Category** – Listed the most purchased products within each category.

Rank	Category	Item Purchased	Total Orders
1	Accessories	Jewelry	171
2	Accessories	Belt	161
3	Accessories	Sunglasses	161
1	Clothing	Blouse	171
2	Clothing	Pants	171
3	Clothing	Shirt	169
1	Footwear	Sandals	160
2	Footwear	Shoes	150
3	Footwear	Sneakers	145
1	Outerwear	Jacket	163
2	Outerwear	Coat	161

9. **Repeat Buyers & Subscriptions** – Checked whether customers with more than 5 previous purchases are more likely to subscribe.

Repeat Buyer (>5 purchases)	Subscribed: No	Subscribed: Yes
False	329	95
True	2,518	958

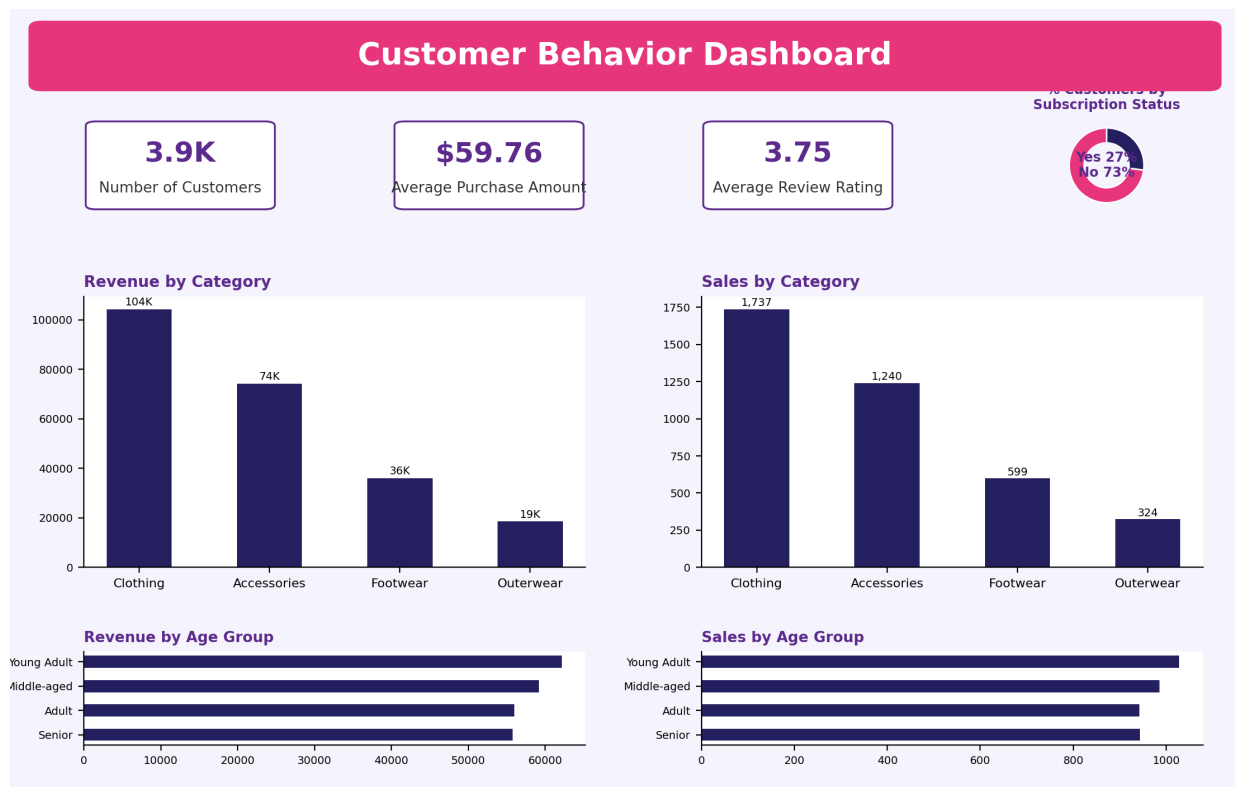
Repeat buyers subscribe at roughly the same 27–28% rate as non-repeat buyers, suggesting purchase frequency alone does not strongly predict subscription.

10. **Revenue by Age Group** – Calculated total revenue contribution of each age group.

Age Group	Total Revenue (USD)
Young Adult (18–31)	62,143
Middle-aged (44–57)	59,197
Adult (31–44)	55,978
Senior (57–70)	55,763

5. Dashboard

Finally, we built an interactive dashboard to present these insights visually, summarizing customer counts, spend, ratings, subscription mix, and revenue/sales breakdowns by category and age group.



6. Business Recommendations

- **Boost Subscriptions** – Promote exclusive benefits for subscribers, since subscribers currently make up only 27% of customers.
- **Customer Loyalty Programs** – Reward repeat buyers to help move the 701 “Returning” customers into the “Loyal” segment.
- **Review Discount Policy** – Balance sales boosts with margin control, particularly for Hats, Sneakers, and Coats, which rely most heavily on discounting.

- **Product Positioning** – Highlight top-rated and best-selling products, such as Gloves, Sandals, Jewelry, and Blouses, in marketing campaigns.
- **Targeted Marketing** – Focus efforts on high-revenue age groups (Young Adults) and faster shipping options (2-Day and Express), which show the highest average purchase amounts.